CS F429: Paper Presentations

1. Overview

Each team selects one paper from the Reading List below. The presentation has two required components:

- **Slides:** a clear presentation covering motivation, method, and key results, plus a critical discussion section: the paper's limitations, open problems it leaves unresolved, and possible future work. It should not just be a summary of what the paper claims.
- **Reproduction, Demo & Viva:** a working implementation of some meaningful part of the paper, run live in front of the instructor with the output tied back to the paper's claims, followed by a viva where the team answers questions on what they simplified, why, and how their results compare to the paper's. This does not need to be full-scale retraining. A scoped-down version (smaller model, smaller dataset, fewer steps) that faithfully demonstrates the core mechanism is expected and encouraged.

2. **Deadline**: No need to submit anything on LMS. Paper presentations will be held with Project Phase 3 demos.

3. Reading List

Each paper below is numbered #1–#27 for easy reference when signing up (e.g., "Group 4 — Paper #17").

#	Title	Authors / Venue
1	**Label Words are Anchors: An Information Flow Perspective for Understanding In-Context Learning**	*Wang, Li, Dai, Chen, Zhou, Meng, Zhou, Sun — EMNLP 2023 Best Paper*
2	**Backward Lens: Projecting Language Model Gradients into the Vocabulary Space**	*Katz, Belinkov, Geva, Wolf — EMNLP 2024 Best Paper*
3	**Text Embeddings Reveal (Almost) As Much As Text**	*Morris, Kuleshov, Shmatikov, Rush — EMNLP 2023 Outstanding Paper*
4	**Entity Tracking in Language Models**	*Kim & Schuster — ACL 2023 Area Chair Award*
5	**Minding Language Models' (Lack of) Theory of Mind: A Plug-and-Play Multi-Character Belief Tracker (SymbolicToM)**	*Sclar, Kumar, West, Suhr, Choi, Tsvetkov — ACL 2023 Outstanding Paper*
6	**Faster Minimum Bayes Risk Decoding with Confidence-based Pruning**	*Cheng & Vlachos — EMNLP 2023 Best Paper*
7	**Look-back Decoding for Open-Ended Text Generation**	*Xu, Zhou, Celikyilmaz, Ma — EMNLP 2023 Outstanding Paper*
8	**LINC: A Neurosymbolic Approach for Logical Reasoning by Combining Language Models with First-Order Logic Provers**	*Olausson, Gu, Lipkin, Zhang, Solar-Lezama, Tenenbaum, Levy — EMNLP 2023 Outstanding Paper*

#	Title	Authors / Venue
9	**SCOTT: Self-Consistent Chain-of-Thought Distillation**	*Wang, Wang, Li, Gao, Yin, Ren — ACL 2023 Outstanding Paper*
10	**Do CoNLL-2003 Named Entity Taggers Still Work Well in 2023?**	*Liu & Ritter — ACL 2023 Reproduction Award*
11	**Ignore This Title and HackAPrompt: Exposing Systemic Vulnerabilities of LLMs Through a Global Prompt Hacking Competition**	*Schulhoff et al. — EMNLP 2023 Best Paper*
12	**Extrinsic Evaluation of Machine Translation Metrics**	*Moghe, Sherborne, Steedman, Birch — ACL 2023 Outstanding Paper*
13	**Generative or Discriminative? Revisiting Text Classification in the Era of Transformers**	*Kasa et al. — EMNLP 2025 Outstanding Paper*
14	**Randomly Removing 50% of Dimensions in Text Embeddings has Minimal Impact on Retrieval and Classification Tasks**	*Takeshita, Takeshita, Ruffinelli, Ponzetto — EMNLP 2025 People's Choice Award*
15	**Linear Classifier: An Often-Forgotten Baseline for Text Classification**	*Lin, Chen, Liu, Lin — ACL 2023 Outstanding Paper*
16	**Marked Personas: Using Natural Language Prompts to Measure Stereotypes in Language Models**	*Cheng, Durmus, Jurafsky — ACL 2023 Social Impact Award*
17	**What the Harm? Quantifying the Tangible Impact of Gender Bias in Machine Translation with a Human-Centered Study**	*Savoldi, Papi, Negri, Guerberof-Arenas, Bentivogli — EMNLP 2024 Social Impact Award*
18	**Mission: Impossible Language Models**	*Kallini, Papadimitriou, Futrell, Mahowald, Potts — ACL 2024 Best Paper*
19	**Hexatagging: Projective Dependency Parsing as Tagging**	*Amini, Liu, Cotterell — ACL 2023 Outstanding Paper*
20	**Revisiting the Optimality of Word Lengths**	*Pimentel, Meister, Wilcox, Mahowald, Cotterell — EMNLP 2023 Outstanding Paper*
21	**Cross-Lingual Consistency of Factual Knowledge in Multilingual Language Models**	*Qi, Fernández, Bisazza — EMNLP 2023 Outstanding Paper*
22	**Infini-gram mini: Exact n-gram Search at the Internet Scale with FM-Index**	*Xu, Liu, Choi, Smith, Hajishirzi — EMNLP 2025 Best Paper*

#	Title	Authors / Venue
23	**Pretraining Data Detection for Large Language Models: A Divergence-based Calibration Method**	*Zhang, Zhang, Guo, de Rijke, Fan, Cheng — EMNLP 2024 Best Paper*
24	**HALoGEN: Fantastic LLM Hallucinations and Where to Find Them**	*Ravichander, Ghela, Wadden, Choi — ACL 2025 Outstanding Paper*
25	**Backpack Language Models**	*Hewitt, Thickstun, Manning, Liang — ACL 2023 Outstanding Paper*
26	**GistScore: Learning Better Representations for In-Context Example Selection with Gist Bottlenecks**	*Gupta, Rosenbaum, Elenberg – ICML 2024*
27	**Coverage-based Example Selection for In-Context Learning**	*Gupta, Gardner, Singh – EMNLP Findings 2023*

4. Grading Rubric (20 points total)

Paper Understanding & Presentation (5 pts)

- Clear explanation of the paper's motivation, method, and key results. (1)
- Correct identification of the paper's core technical contribution. (2)
- Critical discussion of the paper's limitations, open problems, and plausible future work, going beyond limitations the paper itself states. (1)
- Well-organized and visually clear slides. (1)

Technical Depth (4 pts)

- Correct explanation of the core algorithm, architecture, or proof. (2)
- Understanding shown by explaining *why* the method works, not just *what* it does. (1)
- Baselines compared with. (1)

Reproduction, Demo & Viva (11 pts)

- Working, runnable implementation of a meaningful part of the paper (expected more than calling a pretrained checkpoint or a bare API alone). (4)
- Clearly documented and justified scope reduction. (1)
- Real outputs or metrics shown and tied back to the paper's claims. (1)
- Comparison of the team's results with the paper's reported results. (1)
- Viva answers on the team's own implementation choices, limitations, and results. (4)